

CO2 - AT1 - Database Design and Connectivity

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Course code: DSA0503

Subject : Query Processing for Data Science

1) Online Shopping Database (10 Marks)

An online shopping company wants to maintain customer and order information in MySQL. Design Customer and Orders tables with appropriate relationships. Write a Python program to connect to the database, create the tables, insert sample records, and display customer names along with the orders they have placed.

Aim

To create an **Online Shopping Database using MySQL and Python** to store customer and order details, establish a relationship between the Customer and Orders tables, insert sample records, and display customer names along with their orders.

Procedure

1. Import the mysql.connector module.
2. Connect Python to the MySQL server.
3. Create a database named ShoppingDB.
4. Create the Customer table with Customer_ID as the **Primary Key**.
5. Create the Orders table with Order_ID as the **Primary Key** and Customer_ID as a **Foreign Key**.
6. Insert sample customer records into the Customer table.
7. Insert sample order records into the Orders table.
8. Use a JOIN query to combine the Customer and Orders tables.
9. Display the customer names along with their order details.
10. Commit the changes and close the database connection.

Code:

```
import mysql.connector  
  
con = mysql.connector.connect(
```

```

    host="localhost",
    user="root",
    password="123456"
)

cur = con.cursor()
cur.execute("CREATE DATABASE IF NOT EXISTS ShoppingDB")
cur.execute("USE ShoppingDB")
print("Database created successfully")
cur.execute("""
CREATE TABLE IF NOT EXISTS Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100),
    Phone VARCHAR(15)
)
""")
print("Customer table created")
cur.execute("""
CREATE TABLE IF NOT EXISTS Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT,
    Product_Name VARCHAR(100) NOT NULL,
    Quantity INT,
    Amount DECIMAL(10,2),
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
)
""")
print("Orders table created")

```

```
customers = [  
    (1, "Arun", "arun@gmail.com", "9876543210"),  
    (2, "Priya", "priya@gmail.com", "9876501234"),  
    (3, "Rahul", "rahul@gmail.com", "9123456780")  
]
```

```
cur.executemany("""  
INSERT IGNORE INTO Customer  
VALUES (%s, %s, %s, %s)  
""", customers)  
con.commit()  
print("Customer records inserted")
```

```
orders = [  
    (101, 1, "Laptop", 1, 65000.00),  
    (102, 1, "Mouse", 2, 1200.00),  
    (103, 2, "Mobile", 1, 25000.00),  
    (104, 3, "Headphones", 1, 3000.00)  
]
```

```
cur.executemany("""  
INSERT IGNORE INTO Orders  
VALUES (%s, %s, %s, %s, %s)  
""", orders)  
con.commit()  
print("Order records inserted")  
cur.execute("""  
SELECT Customer.Customer_Name,  
       Orders.Order_ID,  
       Orders.Product_Name,  
       Orders.Quantity,  
       Orders.Amount
```

```
FROM Customer
JOIN Orders
ON Customer.Customer_ID = Orders.Customer_ID
""")
records = cur.fetchall()
print("\nCustomer Orders:")
print("-----")
for row in records:
    print("Customer:", row[0])
    print("Order ID:", row[1])
    print("Product:", row[2])
    print("Quantity:", row[3])
    print("Amount:", row[4])
    print("-----")
cur.close()
con.close()
print("Database connection closed")
```

Output:

```
= RESTART: C:\Users\nares\Videos\dsa0503\experiments\ass 2.py
Database created successfully
Customer table created
Orders table created
Customer records inserted
Order records inserted

Customer Orders:
-----
Customer: Arun
Order ID: 101
Product: Laptop
Quantity: 1
Amount: 65000.00
-----
Customer: Arun
Order ID: 102
Product: Mouse
Quantity: 2
Amount: 1200.00
-----
Customer: Priya
Order ID: 103
Product: Mobile
Quantity: 1
Amount: 25000.00
-----
Customer: Rahul
Order ID: 104
Product: Headphones
Quantity: 1
Amount: 3000.00
-----
Database connection closed
```

2) Hospital Patient Database (10 Marks)

A hospital requires a database to manage patient information. Design an SQLite database with a Patient table containing suitable fields and constraints. Write a Python program to connect to the database, create the table, insert patient records, update the contact number of a specified patient, delete a discharged patient's record, and display the remaining records.

Experiment 2: Hospital Patient Database – SQLite

Aim

To create a **Hospital Patient Database using SQLite and Python** to store and manage patient information, perform update and delete operations, and display the remaining patient records.

Procedure

1. Import the Python sqlite3 module.
2. Connect to the SQLite database hospital.db.
3. Create a Patient table with suitable fields and constraints.
4. Set Patient_ID as the **Primary Key**.
5. Insert sample patient records into the table.
6. Update the contact number of a specified patient using the UPDATE command.
7. Delete the record of a discharged patient using the DELETE command.
8. Retrieve the remaining patient records using the SELECT command.
9. Display the patient details.
10. Commit all changes and close the SQLite database connection.

code:

```
import sqlite3

con = sqlite3.connect("hospital.db")

cur = con.cursor()

print("Connected to Hospital Database")

cur.execute("""

CREATE TABLE IF NOT EXISTS Patient (

    Patient_ID INTEGER PRIMARY KEY,

    Patient_Name TEXT NOT NULL,

    Age INTEGER CHECK(Age > 0),
```

```

Gender TEXT,
Disease TEXT,
Contact_Number TEXT UNIQUE,
Status TEXT
)
)
con.commit()
print("Patient table created")
patients = [
    (1, "Arun", 35, "Male", "Fever", "9876543210", "Admitted"),
    (2, "Priya", 28, "Female", "Migraine", "9876501234", "Admitted"),
    (3, "Rahul", 45, "Male", "Diabetes", "9123456780", "Discharged"),
    (4, "Anjali", 32, "Female", "Asthma", "9988776655", "Admitted")
]
cur.executemany("""
INSERT OR IGNORE INTO Patient
VALUES (?, ?, ?, ?, ?, ?, ?)
""", patients)
con.commit()
print("Patient records inserted")
cur.execute("""
UPDATE Patient
SET Contact_Number = ?
WHERE Patient_ID = ?
""", ("9000012345", 2))
con.commit()
print("Patient contact number updated")
cur.execute("""
DELETE FROM Patient

```

```
WHERE Status = 'Discharged'

""")

con.commit()

print("Discharged patient deleted")

cur.execute("SELECT * FROM Patient")

records = cur.fetchall()

print("\nRemaining Patient Records:")

print("-----")

for row in records:

    print("Patient ID:", row[0])

    print("Name:", row[1])

    print("Age:", row[2])

    print("Gender:", row[3])

    print("Disease:", row[4])

    print("Contact:", row[5])

    print("Status:", row[6])

    print("-----")

cur.close()

con.close()

print("Database connection closed")
```


Output:

```
===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/ass 2(1).py =====
Connected to Hospital Database
Patient table created
Patient records inserted
Patient contact number updated
Discharged patient deleted

Remaining Patient Records:
-----
Patient ID: 1
Name: Arun
Age: 35
Gender: Male
Disease: Fever
Contact: 9876543210
Status: Admitted
-----
Patient ID: 2
Name: Priya
Age: 28
Gender: Female
Disease: Migraine
Contact: 9000012345
Status: Admitted
-----
Patient ID: 4
Name: Anjali
Age: 32
Gender: Female
Disease: Asthma
Contact: 9988776655
Status: Admitted
-----
Database connection closed
```